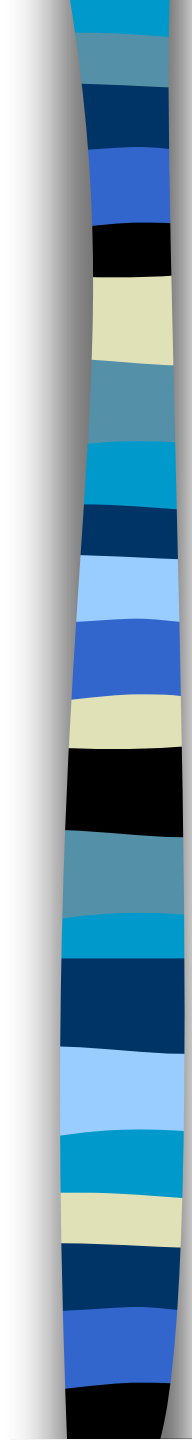


5.1 Database Security :

Introduction to database security, Data Security Requirements, Types of Database Users- Creating, altering and deleting users.

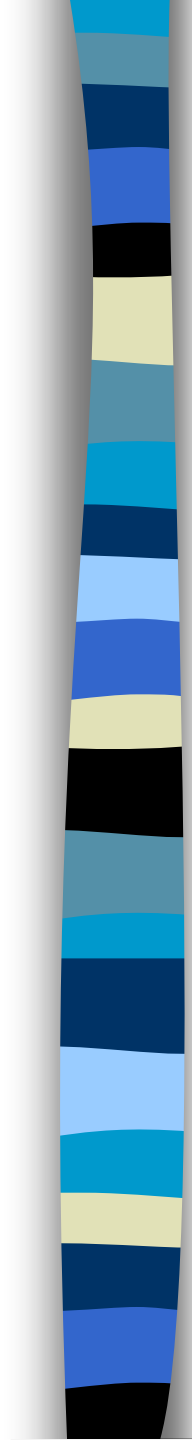


**Prepared By
Mr. Mohammad Ali
Lecturer (Selection Grade)
M. H. Saboo Siddik Polytechnic, Mumbai**



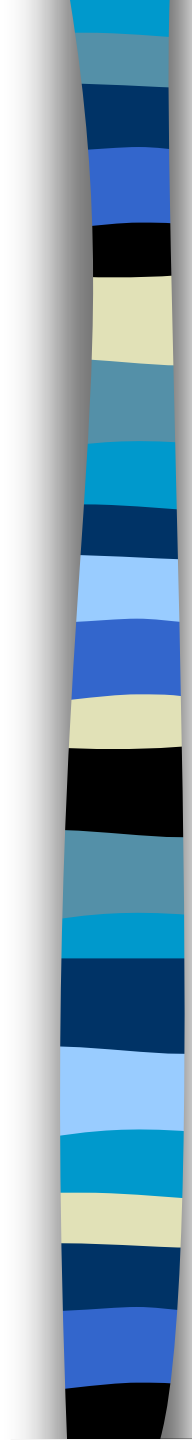
Introduction to database security

- Database security refers to the collective measures used to **protect** and **secure** a **database** from illegitimate use and malicious threats and attacks.
- Database security covers and enforces security on all aspects and components of databases. This includes:
 - Data stored in database
 - Database server
 - Database management system (DBMS)
 - Other database workflow applications
- Database security is generally planned, implemented and maintained by a database administrator and or other information security professional



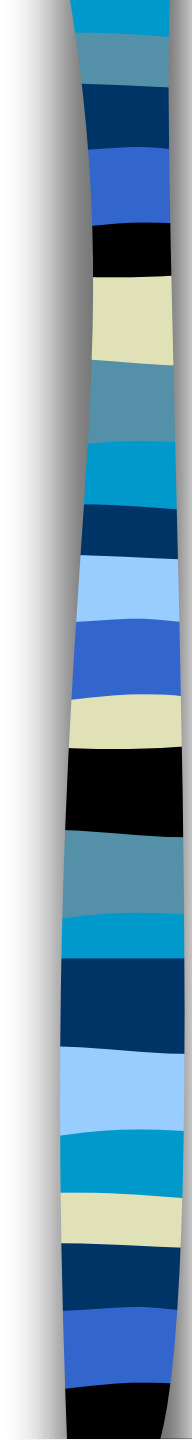
Data Security Requirements

- Database security requirements arise from the need to protect data:
 - first, from accidental loss and corruption, and
 - second, from deliberate unauthorized attempts to access or alter that data.
- Secondary concerns include protecting against undue delays in accessing or using data, or even against interference to the point of denial of service.
- The global costs of such security breaches run to billions of dollars annually.



Data Security Requirements

- These requirements are dynamic.
 - New technologies and practices continually provide new arenas for unauthorized exploitation, as well as new ways for accidental or deliberate misuse to affect even stable products and environments.
 - As security requirements are understood with increasing clarity, certain general principles can be developed for satisfying them and for disabling the threats against them deriving from Internet vulnerabilities.

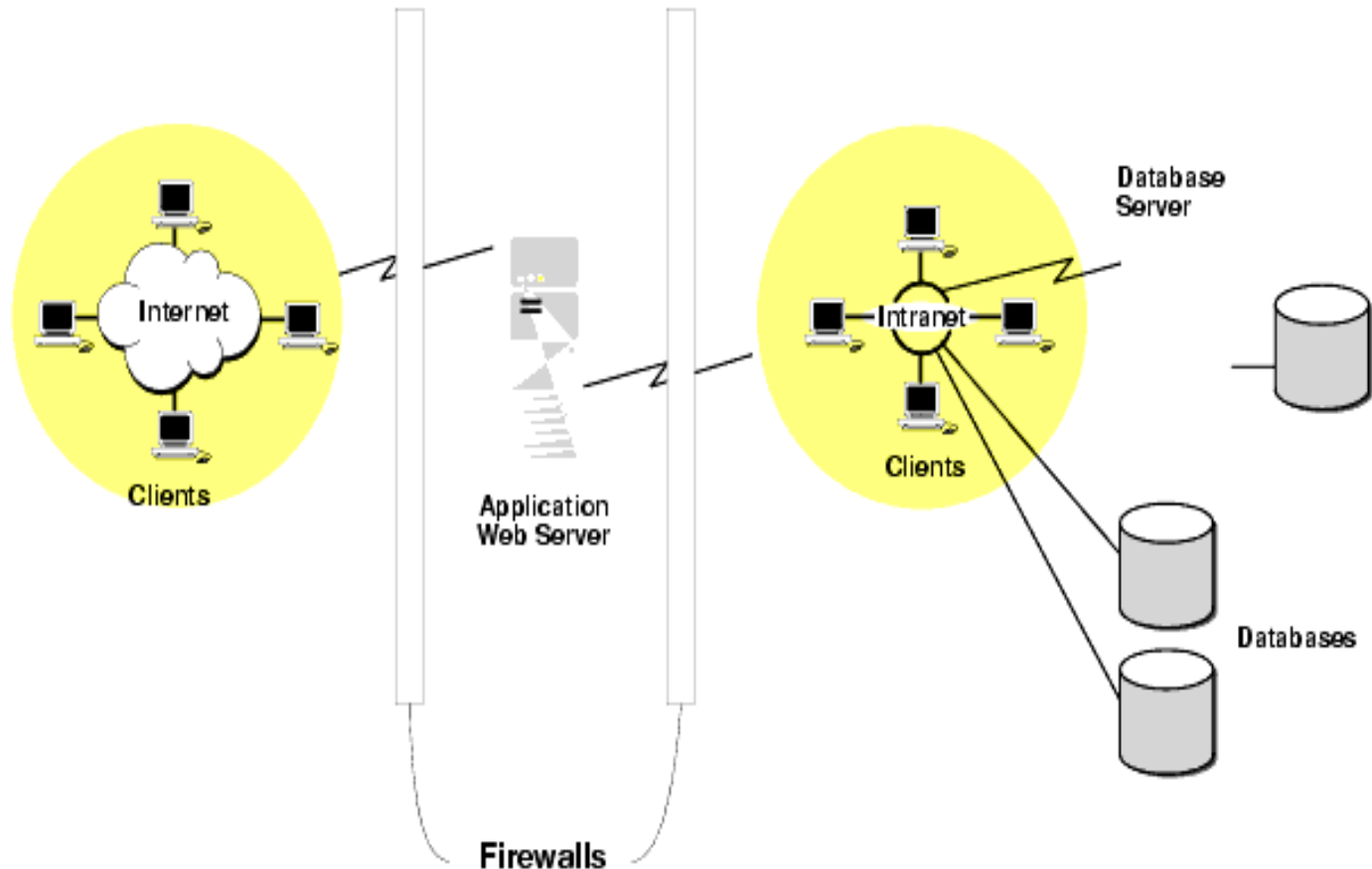


Data Security Requirements

- The basic elements and operations of the database environment include connection to a server or a schema, table access and alteration, and application usage.
 - Securing these against accidental or deliberate misuse is the responsibility of security officers, administrators, and application programmers.
 - They must also administer and protect the rights of internal database users, and guarantee electronic commerce confidentiality as customers access databases from anywhere the Internet reaches.

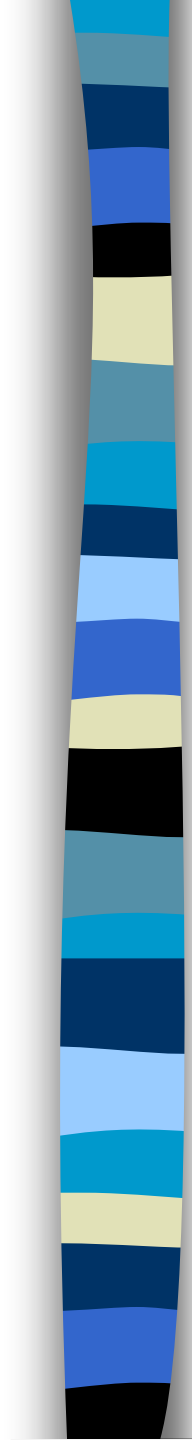
Data Security Requirements

In the Internet age, the full spectrum of risks to valuable and sensitive data, and to user access and confidentiality, is broader than ever before.



Security Issues by Category

Dimension	Security Issues
Physical	Computers must be made physically inaccessible to unauthorized users by keeping them in a secure physical environment.
Personnel	The people responsible for the site's physical security, system administration, and data security must be reliable.
Procedural	The procedures and policies used in the operation of the system must assure reliable data. It is often wise to separate out users' functional roles in data management. For example, one person might be responsible for database backups. His/Her only role is to be sure the database is up and running. Another person might be responsible for generating application reports involving payroll or sales data. His/Her role is to examine the data and verify its integrity. Further, we can establish policies that protect tables and schemas against unauthorized, accidental, or malicious usage.
Technical	Storage, access, manipulation, and transmission of data must be safeguarded by technology that enforces our particular information control policies.



Operating System Authentication

- Many databases (including Microsoft SQL Server 2000) depend on OS to authenticate users
- Once an intruder is inside the OS, it is easier to access the database
- Centralize administration of users
- Ideally, users must be authenticated at each level

Operating System Authentication (continued)

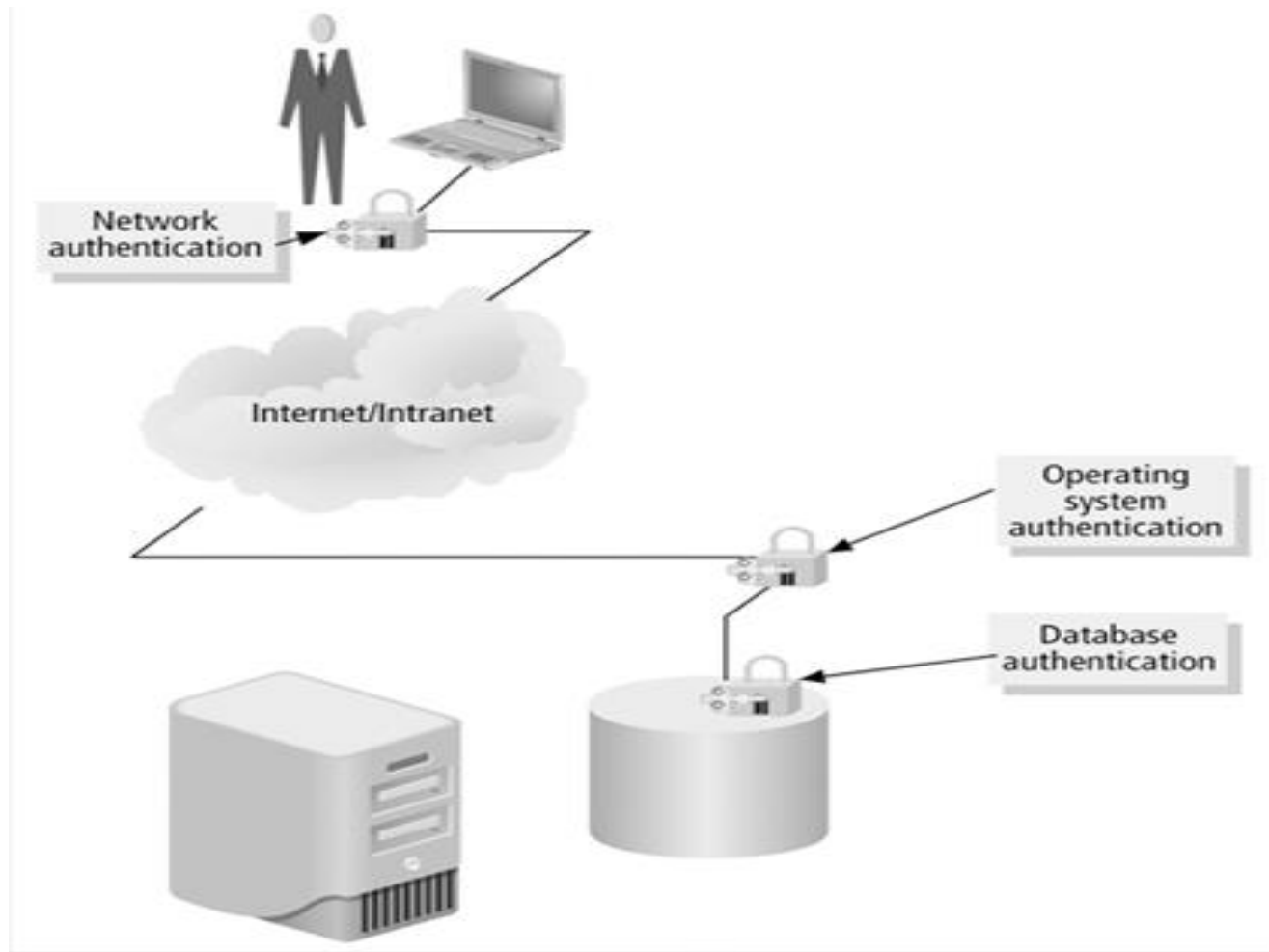
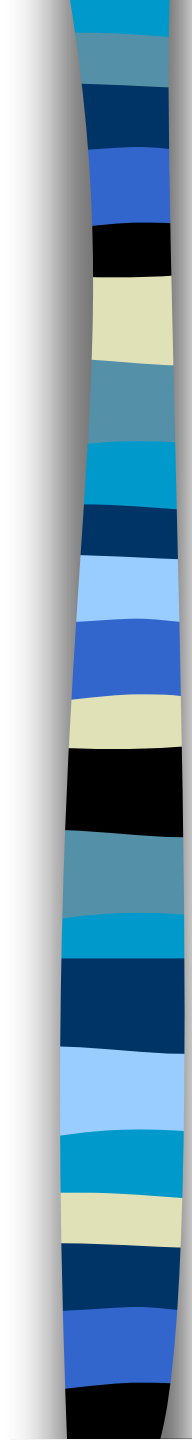
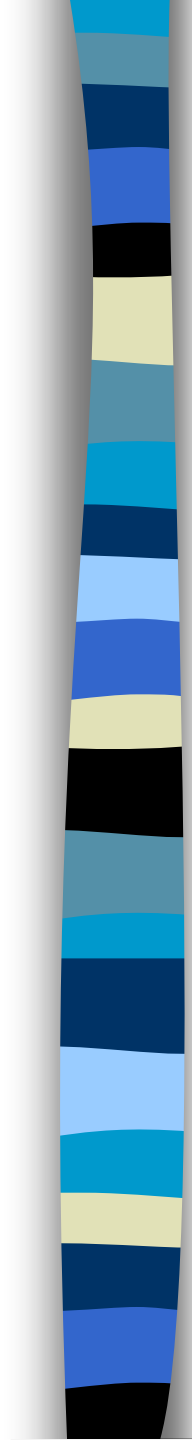


FIGURE Ideal authentication levels for database applications



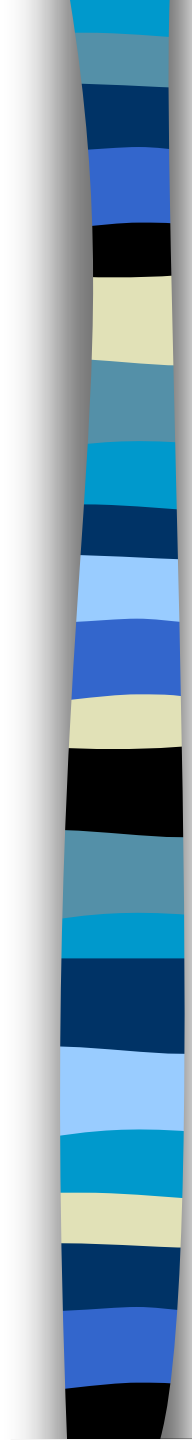
Types of Oracle Database Users

- Database Administrators
- System Analyst
- Application Developers / Programmers
- Application Administrators
- Security Officers
- Network Administrators
- Database Users
 - **Sophisticated Users**
 - **Naive / Parametric End Users**
 - **Casual Users / Temporary Users :**



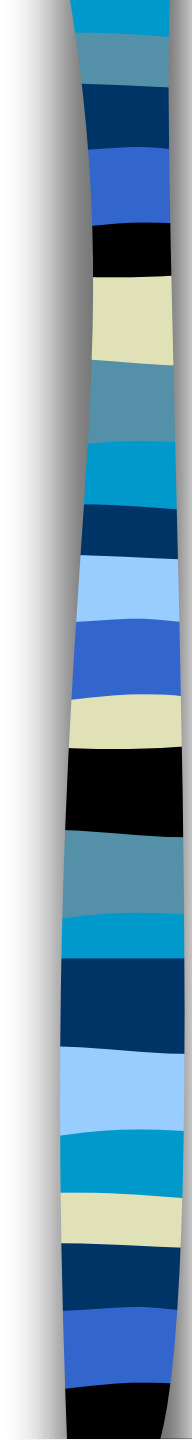
Responsibilities of DBA

- Installing and upgrading the Oracle Database server and application tools
- Allocating system storage and planning future storage requirements for the database system
- Creating primary database storage structures (tablespaces) after application developers have designed an application
- Creating primary objects (tables, views, indexes) once application developers have designed an application
- Modifying the database structure, as necessary, from information given by application developers
- Enrolling users and maintaining system security
- Ensuring compliance with Oracle license agreements
- Controlling and monitoring user access to the database
- Monitoring and optimizing the performance of the database
- Planning for backup and recovery of database information
- Maintaining archived data on tape
- Backing up and restoring the database
- Contacting Oracle for technical support



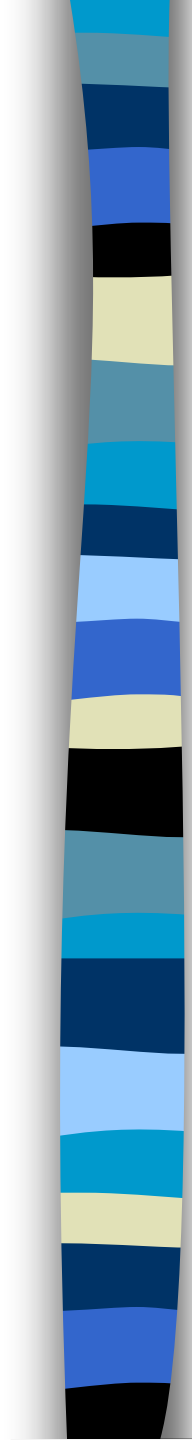
Responsibilities of System Analyst

- Analyzes the requirements of parametric end users.
- Check whether all the requirements of end users are satisfied.



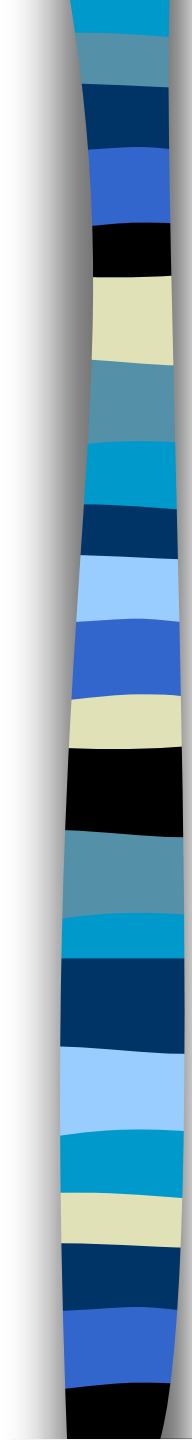
Responsibilities of Application Developers

- Designing and developing the database application
- Designing the database structure for an application
- Estimating storage requirements for an application
- Specifying modifications of the database structure for an application
- Relaying this information to a database administrator
- Tuning the application during development
- Establishing security measures for an application during development



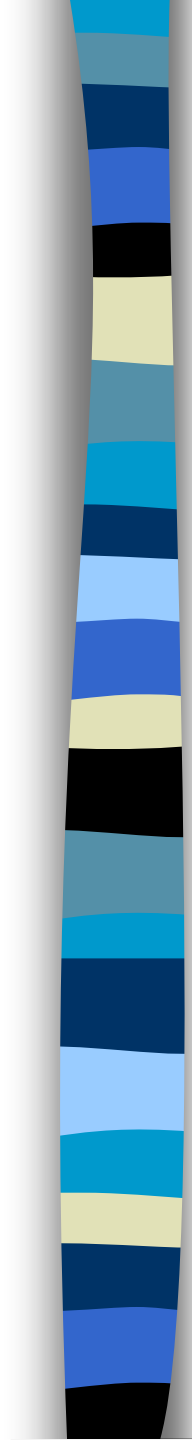
Responsibilities of Application Administrators

- An Oracle Database site can assign one or more application administrators to administer a particular application.
- Each application can have its own administrator.



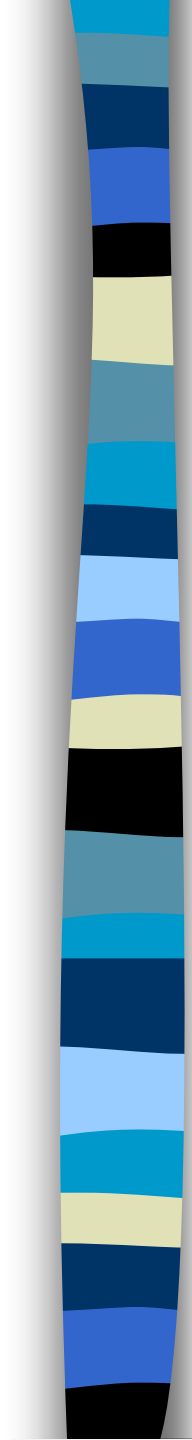
Responsibilities of Security Officers

- A security officer enrolls users, controls and monitors user access to the database, and maintains system security.
- A DBA might not be responsible for these duties if the site has a separate security officer.



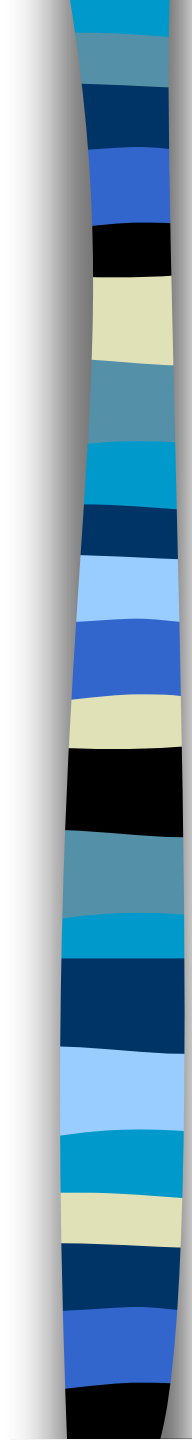
Responsibilities of Network Administrators

- Some sites have one or more network administrators.
- A network administrator, for example, administers Oracle networking products, such as Oracle Net Services.



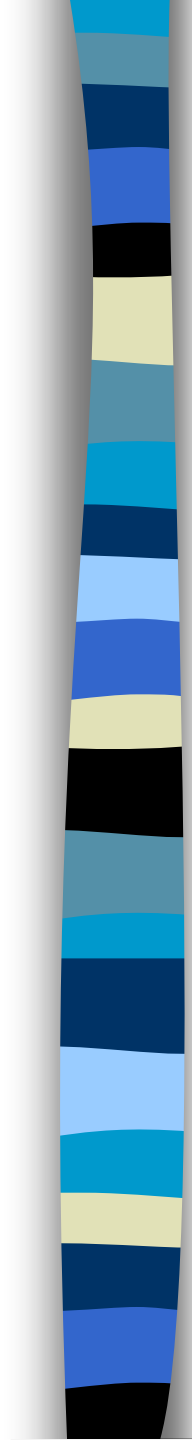
Sophisticated users

- Sophisticated users can be engineers, scientists, business analyst, who are familiar with the database.
- They don't write the program code but they interact the data base by writing SQL queries directly through the query processor.



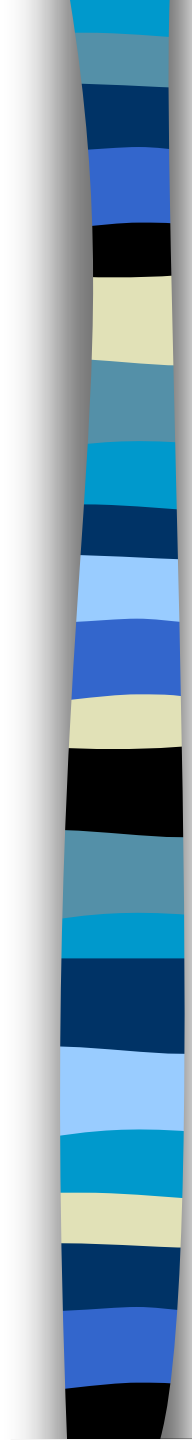
Naive / Parametric End Users

- Parametric End Users are the unsophisticated who don't have any DBMS knowledge but they frequently use the data base applications in their daily life to get the desired results.
- For examples, Railway's ticket booking users are naive users. Clerks in any bank is a naive user because they don't have any DBMS knowledge but they still use the database and perform their given task.



Casual Users / Temporary Users

- Casual Users are the users who occasionally use/access the data base but each time when they access the data base they require the new information, for example, Middle or higher level manager.



Creating Users

- Must be a standardized, well-documented, and securely managed process
- Several ways in Oracle:
 1. **CREATE USER** Statement from **SQL Plus**
 2. **Oracle Enterprise Manager**: GUI administration tool using database authentication
 3. Creating an Oracle User Using External (Operating System) Authentication
 4. **SQL developer**

Creating Users

- In Oracle, use the **CREATE USER** statement:
 - Part of the a Data Definition Language (DDL)

```
CREATE USER {name}  
IDENTIFIED {BY password | EXTERNALLY | GLOBALLY  
as 'external_name'}  
[DEFAULT TABLESPACE {tbspname}]  
[TEMPORARY TABLESPACE {tmpname}]  
[QUOTA {integer {K|M} ON {tbspname}]  
[PROFILE {pname}]  
[PASSWORD EXPIRE]  
[ACCOUNT {lock | unlock}]
```

Creating an Oracle User

- **IDENTIFIED** clause
 - Tells Oracle how to authenticate a user account
 - BY **PASSWORD** option: encrypts and stores an assigned password in the database
 - **EXTERNALLY** option: user is authenticated by the OS
 - **GLOBALLY AS** option: depends on authentication through centralized user management method

Creating an Oracle User (continued)

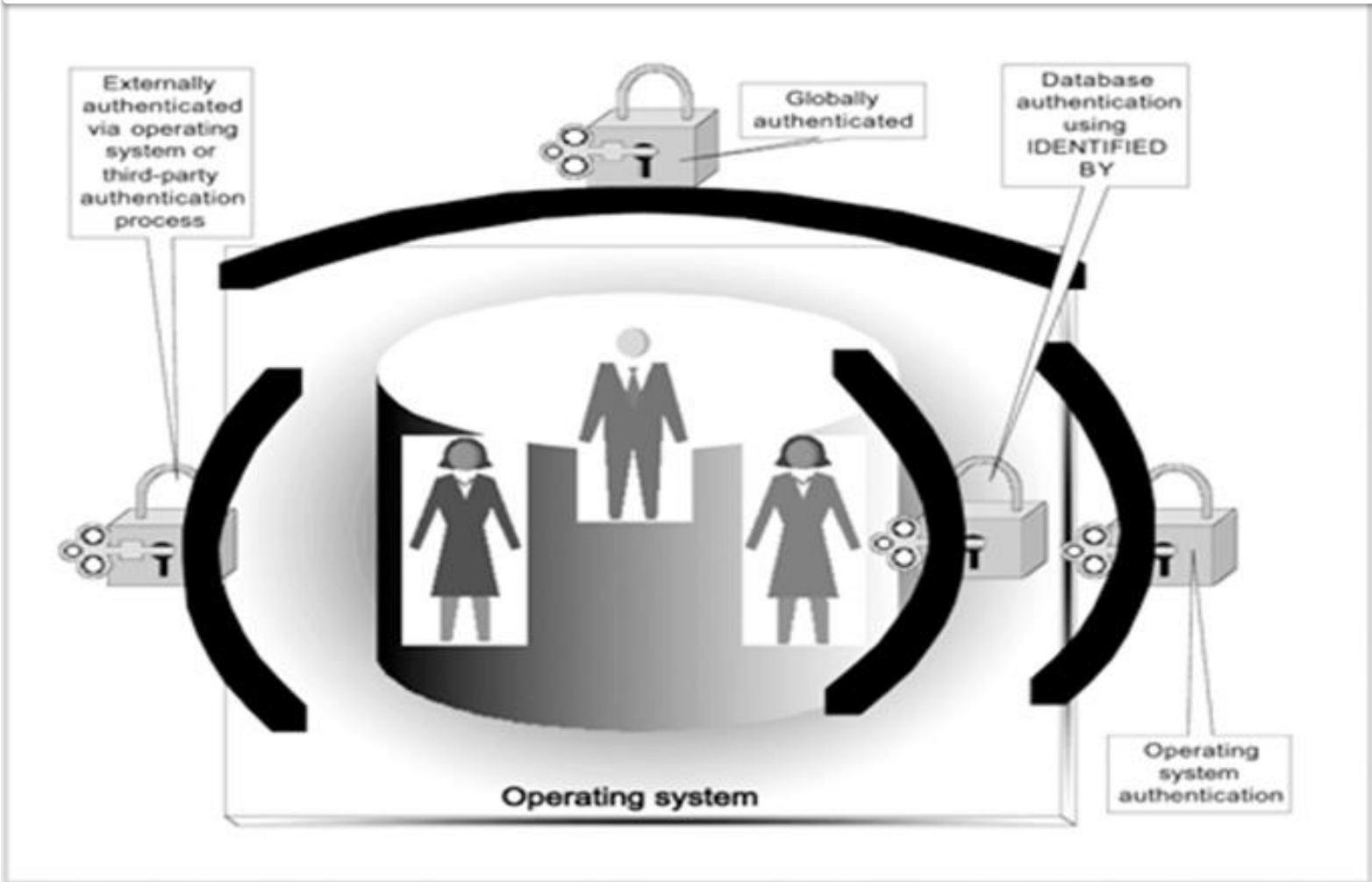
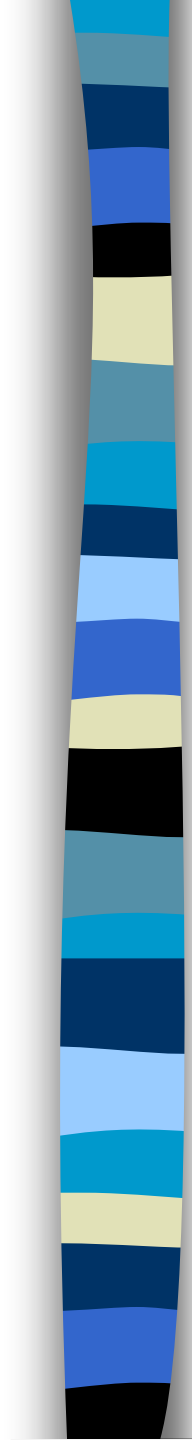


FIGURE Architecture of Oracle authentication methods

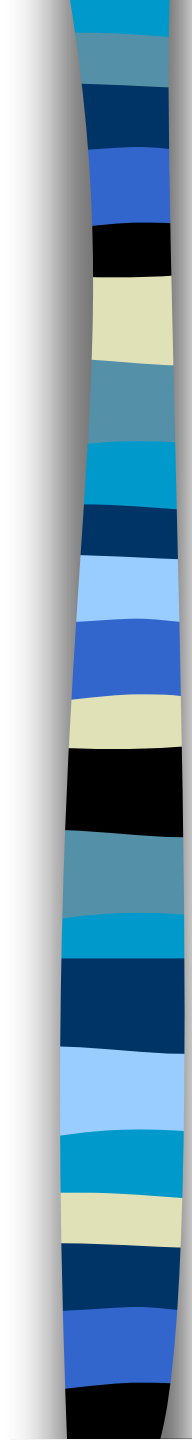


Creating an Oracle User (continued)

- **DEFAULT TABLESPACE** clause: specifies default storage for the user
- **TEMPORARY TABLESPACE** clause
- **QUOTA** clause: tells Oracle DB how much storage space a user is allowed for a specified tablespace
- **PROFILE** clause: indicates the profile used for limiting database resources and enforcing password policies

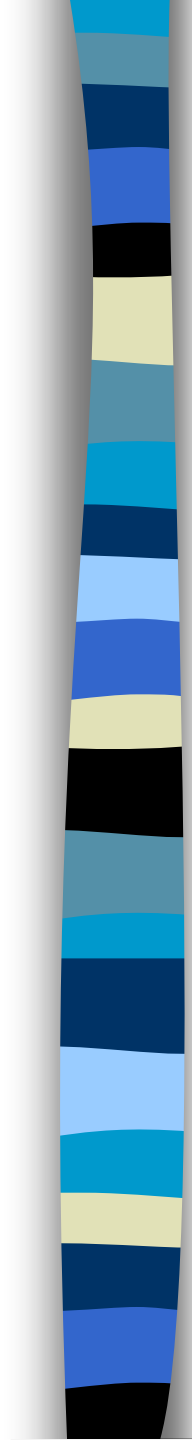
Creating an Oracle User - Example

```
CREATE USER STUDENT  
IDENTIFIED BY 123456  
DEFAULT TABLESPACE USERS  
TEMPORARY TABLESPACE TEMP  
QUOTA 10M ON USERS  
PROFILE DEFAULT  
ACCOUNT UNLOCK  
/
```



Creating an Oracle User (continued)

- **PASSWORD EXPIRE** clause: tells Oracle to expire the user password and prompts the user to enter a new password
- **ACCOUNT** clause: enable or disable account
- **ALTER USER**: modifies a user account



Altering / Modifying Users (Oracle Users)

- Modifications involve:
 - Changing passwords
 - Locking an account
 - Increasing a storage quota
- By using **ALTER USER** statement
 - Part of the a Data Definition Language (DDL)
- By using **Oracle Enterprise Manager**: graphical tool

Altering / Modifying Users (Oracle Users)

- Even DBA can not recover real value of password from the database
- We can change the password and inform the user of the new password
- We make the password expire immediately so the user must choose a new password that he /she finds easier to remember.

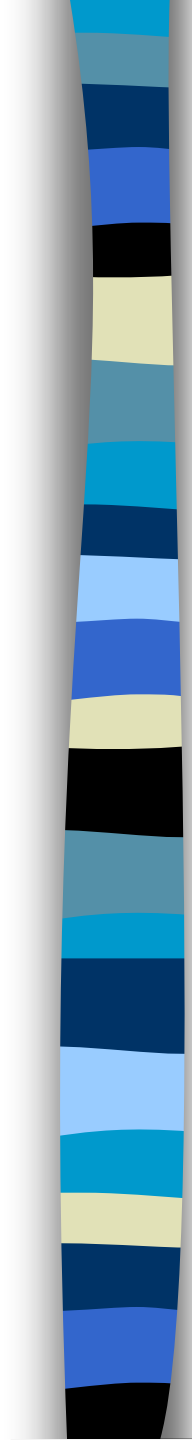
```
ALTER USER STUDENTA  
IDENTIFIED BY STUDENTA  
PASSWORD EXPIRE;
```

Deleting / Removing Users (Oracle Users)

- **DROP** command
- **CASCADE** option: when user owns database objects

DROP USER STUDENT CASCADE;

- Recommendations:
 - Make a backup first. Backup the account for one to three months
 - Listing all owned objects
 - Lock the account or revoke the **CREATE SESSION** privilege
 - Obtain a written request (for auditing purposes)



Default Users

- Oracle default users:
 - SYS, owner of the data dictionary
 - SYSTEM, performs almost all database tasks
 - ORAPWD, creates a password file